

FarmsKing Project Flow & Architecture

1. User Registration & Authentication Flow

- User opens the App/Website.
- Chooses to Login or Register.
- Enters Mobile Number -> OTP Verification (via Firebase/WhatsApp).
- Selects a Role (Farmer, Gardener, Advisor, Customer, Business Partner).
- Completes Profile Setup.
- Redirected to the respective Dashboard based on the Role.

2. Core Modules & Dashboards

- **Farmer Dashboard:** Manage Farms, Crops, Expenses, Market Rates, Ask Agri-AI.
- **Gardener Dashboard:** Manage Plants, Lawns, Watering schedules.
- **Advisor Dashboard:** Receive consultation requests, Video/Audio calls via Agora, Earn consultation fees.
- **Agri Store (Customer):** Browse products (Seeds, Fertilizers), Add to Cart, Checkout via UPI, Track Orders.
- **Business Partner:** Refer users, Track referral income, Manage Wallet, Request Withdrawals.

3. Agri Store E-Commerce Flow

- User browses the store categories (Seeds, Machinery, etc.).
- Clicks on a product -> Views details and reviews.
- Adds product to Cart -> Proceeds to Checkout.
- Enters Delivery Address.
- Payment Gateway (UPI link generation).
- Payment Confirmed -> Order Placed.
- Admin Panel -> Order Processing -> Dispatch -> Delivered.

4. Agri-AI & Expert Consultation Flow

- User faces an issue (e.g., Yellow leaves on crop).
- **Option A (Agri-AI):** Uploads photo -> Gemini AI analyzes -> Gives instant solution.
- **Option B (Expert Advisor):** Books an appointment -> Pays fee -> Video call via Agora -> Receives prescription.

5. Wallet & Referral System Flow

- User signs up with a Referral Code (Business Partner).
- Business Partner gets referral commission credited to Wallet.
- Partner checks Wallet Balance in App.
- Requests Withdrawal (Minimum amount reached).
- Admin approves -> Money transferred to Partner's Bank/UPI.

6. Technical Execution Flow

- **Frontend (React Native / Expo):** Renders UI, handles user clicks, sends API requests.
- **Backend (NestJS):** Receives requests, applies business logic (auth, wallet rules).
- **Database (PostgreSQL via Prisma):** Saves/Retrieves data (Users, Orders, Wallet).
- **Third-Party Services:** Firebase (Auth/Notifications), Agora (Video Calls), Gemini (AI), Open-Meteo (Weather).